

Rational Expressions: Simplify vs Solve

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Standard: CCSS.MATH.CONTENT.HSA-APR.D.6 & HSA-REI.A.2 — Rewrite rational expressions using algebra; solve rational equations and identify extraneous solutions.

⚠ READ FIRST — Simplify vs Solve

SIMPLIFY (Part A): No equals sign. You start with an *expression* and rewrite it in a simpler form. Your answer is another expression. Assume no denominator equals zero — you do **not** check for extraneous values.

SOLVE (Part B): Has an equals sign. You start with an *equation* and find the value(s) of the variable that make it true. Your answer is a number (or numbers). You **must** check every solution and reject any that makes a denominator zero (extraneous).

Part A — SIMPLIFY (no denominator = 0)

Directions: Simplify each rational expression completely. Show all factoring and cancellation steps. Assume no denominator equals zero. Do not solve — there is nothing to solve.

1. $\left(y - \frac{2}{y-1}\right) \left(y + \frac{1}{y-2}\right)$

2. $\frac{x^2 - 9}{x^2 + 5x + 6}$

3. $\frac{3}{x-2} - \frac{2}{x+1}$

4. $\frac{x^2 - 4}{x^2 - x - 6} \div \frac{x^2 + 4x + 4}{x^2 - 9}$

Part B — SOLVE (check extraneous)

Directions: Solve each rational equation. After finding candidate solutions, **check** each one in the original denominators. Reject any value that makes a denominator zero (extraneous). Write your final answer as $x = \dots$

5. $\frac{x}{x - 1} - \frac{2}{x + 1} = \frac{4}{x^2 - 1}$

6. $\frac{1}{x} + \frac{1}{x + 3} = \frac{1}{2}$

7. Work-Rate Word Problem. Two workers can finish painting a house together in 6 hours. Working alone, the faster worker finishes the job 5 hours sooner than the slower worker.

How many hours does **each worker** take to finish the job alone? (*Set up a rational equation using rates = 1 / time, then solve.*)

8.
$$\frac{2x}{x-3} = \frac{6}{x-3} + \frac{x}{2}$$